With 750 labelled tweets, does fine-tuning beat a calibrated open LLM?

RESEARCH QUESTION (UNCHANGED FROM THE PROPOSAL)

If we only have 750 labelled tweets, does fine-tuning BERTurk on them give a higher macro-F1 on the official OffensEval-TR 2020 test set (3,528 tweets) than prompting an open LLM (Gemma-2-9B-it or Qwen3-8B) that uses the same 750 tweets: 8 as examples in its prompt, and the other 742 to choose its decision threshold?

THE ANSWER IS NO IF

against either LLM, BERTurk's average over 10 runs is not at least 0.02 higher than the LLM's average over 3 runs, or a paired bootstrap test says the gap could be chance.

0.763

TF-IDF baseline on the frozen test set (95% CI 0.745–0.779)

60

baseline errors audited, grouped into categories with counts

0.790

Gemma-4-12B few-shot dev macro-F1 after threshold calibration

Vesile Han · Gürkan Göztepeli (CSE 654, research leads) · Mert Uğur · Utku Gökçek (CSE 484)

Talk: 10 slides + 4 Project Card appendix slides (graded, not presented).

30,000 train / 1,756 dev / 3,528 frozen test; 19.3% offensive

File / split	Tweets	NOT	OFF	OFF %
Official training file	31,756	25,625	6,131	19.3
Our train (seed 42)	30,000	24,208	5,792	19.3
Our dev (never tuned on)	1,756	1,417	339	19.3
Official test (frozen)	3,528	2,812	716	20.3

OffensEval-TR 2020 (Çöltekin, LREC 2020). Turkish tweets, annotations CC BY 2.0. The released files already mask @USER and URLs; we add no cleaning. Nine duplicate texts inside train; one test tweet also appears in train (0.03%).

REAL EXAMPLE, LABEL OFF (TRAIN ID 33518)

“Kırmızı ağır karar Allah belanı versin”

“A red card is a harsh call, God damn you.” Swearing at a decision, not at a person: the label is still OFF because the annotation guidelines treat the curse as offensive.

CONTRAST EXAMPLE, LABEL NOT

“...annem alacağı arabadan kaç kere vazgeçti ya kahretsin zalım trafik”

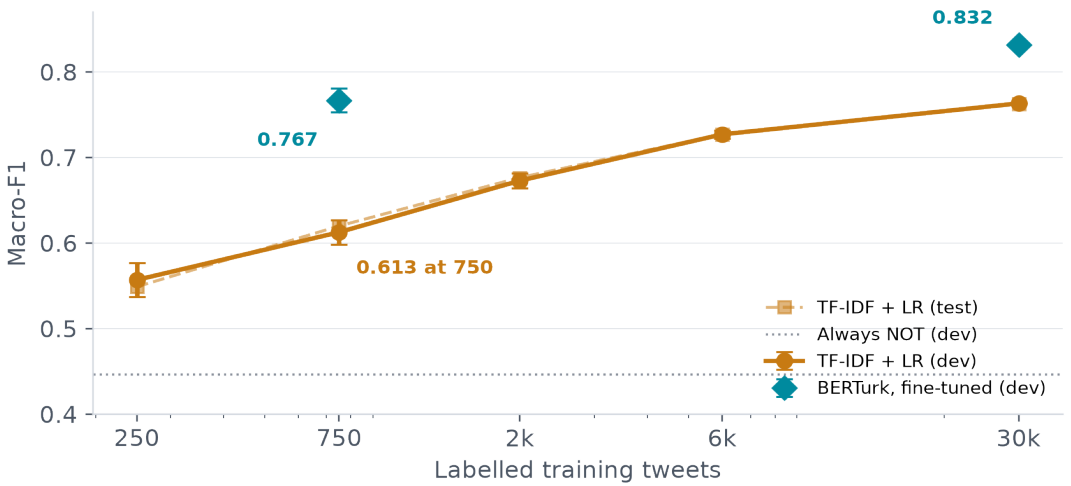
“How many times my mum gave up on buying a car, damn this cruel traffic.” Cursing at traffic is not offensive; target matters.

WHY THESE COUNTS MATTER

Only 19.3% of training tweets are offensive. A model that always says “not offensive” is 80.7% accurate but scores 0.447 macro-F1, so accuracy would hide the failure; we use macro-F1. Rules: test frozen since the split; dev is for code checks and mid-project numbers; no method setting is chosen by a dev score; every system uses the saved subsets (seed 42 + k).

TF-IDF + logistic regression reaches 0.763 macro-F1 on test, but 0.620 with 750 labels

System	Test macro-F1	OFF P	OFF R	Spread
Always “not offensive”	0.444	—	0.000	deterministic
TF-IDF + LR, 750 labels	0.620 ± 0.012	—	—	sd over 10 samples
TF-IDF + LR, 30,000 labels	**0.763**	0.589	0.673	95% CI [0.745, 0.779]



Dev curve (solid) and test curve (dashed) for the same TF-IDF runs. The LLM lines are flat because the LLM always gets 750 labels. BERTurk-750 reaches 0.767 and BERTurk-30k 0.832 on dev.

WHAT THIS MEANS FOR THE RESEARCH QUESTION

At 750 labels, the cheap baseline is 0.620 on test and 0.613 on dev: that is the floor BERTurk and the LLMs must beat. At 30,000 labels the same model reaches 0.763, about 0.05 below the published BERTurk score (0.816, KUISAIL), so the full-data ceiling is not the target of the 750-label question. Fine-tuned BERTurk reaches 0.767 with 750 labels and 0.832 with 30,000.

275 dev errors; a 60-error audit shows where a word model fails

False alarms (166) — category	n
No insult; spurious word/char cue	9
Political or religious criticism	10
Sports criticism or banter	5
Personal criticism not judged offensive	3
Sarcasm or irony	1
Label noise or genuinely ambiguous	2

Missed offensive (109) — category	n
Implicit insult, no profanity	8
Masked or abbreviated profanity	5
Veiled threat or violence	3
Political or religious insult	4
Institutional or sports criticism	3
Label noise or not offensive	7

Counts are in the 60-error audit (30 per direction, seed 42), not over all 275 errors. Both tables therefore sum to 30.

False alarm · F3 · id 13105

“Aziz Yıldırım Fenerbahçe’yi padişahlık zannediyor herhalde…”

Club criticism without a slur. Conflict vocabulary fires.

Miss · M1 · id 43325

“…bundan daha parazit canlı türü yok”

“No more parasitic species”: an insult with no swear word.

Miss · M2 · id 15069

“…bakanlık seçimi kaldı amkk”

Masked profanity (“amkk”) is unseen by the word/char model.

Lesson: false alarms cluster on conflict and club vocabulary and character n-grams; misses cluster on implicit insults and masked profanity. Roughly one in seven audited errors is label noise or a genuinely ambiguous tweet — no threshold can fix that. We will use this audit to choose prompt examples and to report an annotation-noise ceiling.

Both arms run end-to-end; first dev numbers

1. **750 labelled tweets**
   One random sample, shared with every system

2. **8 prompt examples**
   4 offensive, 4 not; fixed per run

3. **742 calibration tweets**
   The LLM scores these too; choose the threshold with best macro-F1

4. **Apply to dev**
   Score = log P(A) − log P(B); A = offensive, B = not

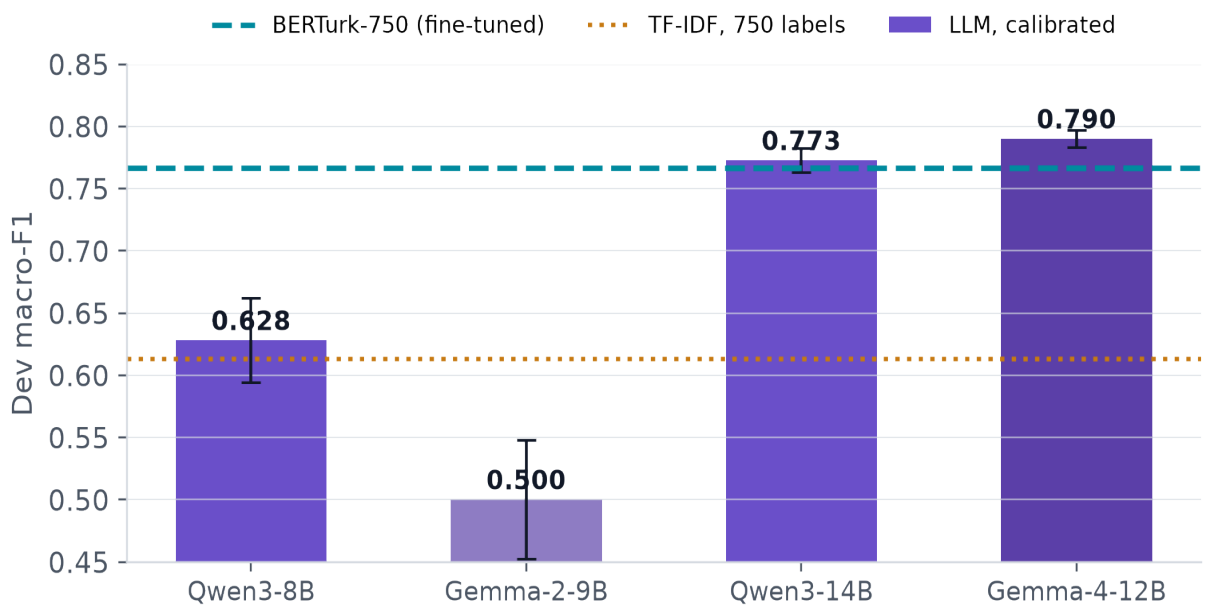
Fine-tuning arm: BERTurk and XLM-R-base, 46 runs on the shared subsets. The four steps above are the LLM arm.

Controlled change: baseline trains weights on the 750 labels; the LLM uses the identical 750 labels to pick 8 examples and a threshold. Nothing else differs except how the labels are used.

System (dev, 1,756 tweets)	Macro-F1	Runs	Threshold
TF-IDF + LR, 750 labels	0.613 ± 0.015	10	—
Qwen3-8B, zero-shot	0.162	1	0
Qwen3-8B, 8 examples + 742	**0.628**	3	3.13, 2.75, 2.50
Gemma-2-9B, zero-shot	0.412	1	0
Gemma-2-9B, 8 examples + 742	**0.500**	3	0.63, 0.25, 0.88
BERTurk, 750 labels	**0.767 ± 0.014**	10	—
BERTurk, 30,000 labels	0.832 ± 0.003	3	—

Gemma-2-9B runs in 4-bit on an L4 with float16-compatible settings; Qwen3-8B runs with thinking mode off. Both use the same Turkish instructions and answer options. BERTurk is fine-tuned with the proposal settings (128 tokens, batch 16, 2e-5, 4 epochs or 400 updates).

BERTurk beats both proposal LLMs; Gemma-4-12B leads on dev



System	Dev macro-F1	Runs
TF-IDF + LR, 750	0.613 ± 0.015	10
Qwen3-8B	0.628 ± 0.034	3
Gemma-2-9B	0.500 ± 0.048	3
Qwen3-14B	0.773 ± 0.010	3
Gemma-4-12B	**0.790 ± 0.007**	3
BERTurk, 750	**0.767 ± 0.014**	10
BERTurk, 30,000	0.832 ± 0.003	3

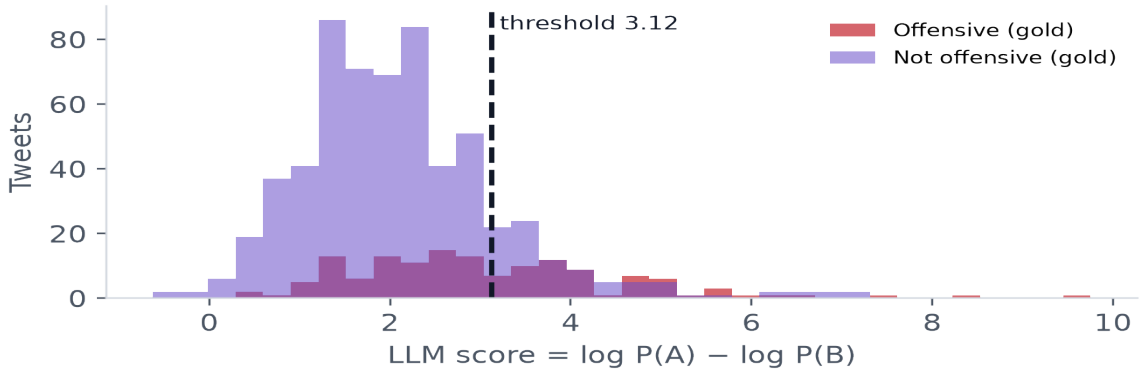
Dev macro-F1, calibrated LLMs. BERTurk is fine-tuned. Bars show ± sd over runs; the dashed line is BERTurk-750.

BERTurk-750 is +0.154 over TF-IDF and +0.139 / +0.267 over Qwen3-8B / Gemma-2-9B. Gemma-4-12B is +0.023 over BERTurk-750. Test-set comparison is pending; all numbers are dev.

Qwen’s scores are shifted; Gemma’s threshold is already near optimal

Zero-shot diagnostic	F1 @ 0	F1 @ best	AUC	mean OFF	mean NOT
Qwen3-14B	0.171	0.677	0.756	2.79	2.10
Gemma-4-12B	0.763	0.765	0.864	1.03	-4.13

Dev, 1,756 tweets. F1 @ 0 uses the zero-shot threshold; F1 @ best is the best possible threshold on the same scores. mean OFF / mean NOT are mean log P(A) − log P(B) by gold class.



One Qwen3-8B run: the chosen threshold sits between the two gold classes. The same 742 labels are the only labels used to set it.

WHAT EXPLAINS THE GAP

Not the tokenizer. Gemma is more compact on these tweets (32.3 vs 37.9 tokens/tweet), yet Qwen calibrates better.

Qwen is shifted, not blind. At threshold 0 it calls 99% of tweets offensive, but the ranking is usable: best-threshold F1 is 0.677.

Gemma is already centred. Its F1 at 0 is 0.763 and its best is 0.765, so the 742 labels can only add +0.027.

The 8 examples still lift Qwen from 0.677 to 0.773, so calibration is not the whole gain.

Threshold geometry, not proof of contamination. The exact-completion check is still planned.

One dev pass on an L4: 3–5 tweets/s with examples, 10–18 without

System	Params	Dev scoring	Throughput	Peak VRAM	Notes
TF-IDF, 750	—	0.6 s	~3,100/s	CPU	fit 1.0 s
TF-IDF, 30,000	—	0.6 s	~3,100/s	CPU	fit 14.6 s
Qwen3-8B, few-shot	8.2B	355 s	5.0/s	—	3 runs
Gemma-2-9B, few-shot	9.2B	430 s	4.1/s	—	3 runs
Qwen3-14B, few-shot	14.8B	644 s	2.7/s	—	3 runs
Gemma-4-12B, few-shot	12.0B	579 s	3.0/s	—	3 runs
Qwen3-14B, zero-shot	14.8B	180 s	9.8/s	10.7 GB	load 157 s · 1,290 tok/s

All times are for scoring the 1,756 dev tweets; model download and load are excluded unless stated. Few-shot throughput is 3–4× lower than zero-shot because each prompt adds 8 examples (~600 tokens).

Measured on one L4, 4-bit NF4, batch 6–8. Qwen3-14B zero-shot: load 157 s, 1,290 prompt tokens/s, peak 10.7 GB. Gemma-4-26B-A4B exceeded 22 GB and could not load. Reported runs: 8.8 Colab units (~2.3 L4-hours); 9.7 including probes. BERTurk: 35 s training at 750 labels, ~29 min at 30,000.

BERTurk and Ablation A are done; remaining steps have a date

Date	Experiment	Owner	Status
12 Nov	Mid-project talk; freeze this deck	TBD	today
13–16 Nov	BERTurk 750 (10 runs) and 30,000 (3 runs) on dev	TBD	done
17–20 Nov	Ablation A: BERTurk vs XLM-R-base, 750 labels, 10 runs	TBD	done
21–24 Nov	Ablation B: Gemma with English instructions, 3 runs	TBD	planned
25–27 Nov	Test-contamination check on the first 200 test tweets	TBD	planned
28–30 Nov	Freeze methods; open test; final runs for every system	TBD	planned
1–5 Dec	Paired bootstrap, final error analysis, paper draft	TBD	planned
8 Dec	Final report (short research paper)	TBD	planned

BIGGEST RISK

A 26B+ model may not fit the L4: Gemma 4 26B-A4B OOM'd at 22 GB, so we used the 12B dense model. Fallback: Qwen3-14B alone. Every run writes one CSV and a JSON, so a lost session costs one run, not the experiment.

RESULT RISK

The lines may never cross. The 750-label question still has a clear answer; the learning curve is descriptive, with no significance test at the other sizes.

Response to proposal feedback: the rubric asks for baseline numbers on the frozen test set, so we scored the deterministic TF-IDF baseline once on test and kept all method results on dev. The research question and the 0.02 / bootstrap rules are unchanged.

What we know, what we do not, and what comes next

0.763

test macro-F1 for the TF-IDF baseline at 30,000 labels; 95% CI [0.745, 0.779]. At 750 labels it falls to 0.620 ± 0.012.

0.790

Gemma-4-12B dev macro-F1 after calibration; zero-shot is 0.763. Qwen3-14B reaches 0.773.

0.767

BERTurk-750 dev macro-F1 (10 runs); 30,000 labels reaches 0.832 ± 0.003. It beats Qwen3-8B and Gemma-2-9B, but Gemma-4-12B (0.790) still leads on dev.

The mid-project answer is a baseline and a working method, not the final verdict. What we do not have yet: any test-set method result. BERTurk-750 is 0.767 on dev; the final answer uses test. What is fixed: the frozen test set, the metric, the 0.02 threshold and the paired bootstrap.

Questions welcome — any member can answer any part.

Fields 1–3: title and abstract, research question, contributions

1 · TITLE AND ABSTRACT

Working title: Fine-tuning BERTurk vs threshold-calibrated open LLMs for Turkish offensive language with 750 labels

Small Turkish platforms can label only a few hundred tweets. We ask whether fine-tuning BERTurk on 750 labelled OffensEval-TR tweets beats an open LLM that uses the same 750 labels as 8 prompt examples and 742 calibration tweets for its decision threshold. The TF-IDF + logistic-regression baseline scores 0.763 macro-F1 on the frozen 3,528-tweet test set, but 0.620 with 750 labels. The LLM arm scores each tweet as log P(A) − log P(B); the 742 labels set its threshold. On dev, calibration moves Qwen3-8B from 0.162 to 0.628 and Gemma-2-9B from 0.412 to 0.500; Qwen3-14B reaches 0.773 (zero-shot 0.171) and Gemma-4-12B 0.790 ± 0.007 (zero-shot 0.763). Fine-tuned BERTurk reaches 0.767 ± 0.014 on dev with 750 labels. [GAP: BERTurk-750 vs LLM test result, after the freeze.] The final paper reports the paired bootstrap gap and the label count at which fine-tuning overtakes the LLM line.

2 · RESEARCH QUESTION

Unchanged from the proposal. With 750 labelled tweets, does fine-tuning BERTurk beat prompting Gemma-2-9B-it or Qwen3-8B that uses the same 750 tweets (8 examples, 742 threshold labels), on macro-F1 over the official 3,528-tweet test set?

“No” result: against either LLM, BERTurk’s mean over 10 runs is not ≥0.02 higher than the LLM’s mean over 3 runs, or the paired bootstrap says the gap could be chance ($p \geq 0.05$).

Change: none. The data and method forced no change; the mid-project rubric added a test-set baseline, which is reported separately and does not alter the question.

3 · CONTRIBUTIONS (TESTABLE CLAIMS)

C1. At 750 labels, BERTurk’s mean test macro-F1 is at least 0.02 above each LLM’s, and a paired bootstrap rejects chance. This is the core claim and is awaiting the frozen-test comparison; on dev BERTurk (0.767) is above Qwen3-8B (0.628) and Gemma-2-9B (0.500).

C2. Choosing the threshold on the 742 non-example labels improves each LLM’s dev macro-F1 over zero-shot by a measurable margin; we report the per-model delta and its run-to-run spread.

C3. The learning curve places the label count where fine-tuning overtakes the LLM line between 250 and 6,000; if the lines never cross, the 750-label answer still stands.

Fields 4–5: related work and data

4 · RELATED WORK (5–8 REFERENCES; ONE LINE ON THE DIFFERENCE)

Zampieri et al. (2020). SemEval-2020 Task 12: Multilingual Offensive Language Identification in Social Media (OffensEval 2020).

Top Turkish systems fine-tune transformers on the full training data; no label-budget comparison.

Çöltekin (2020). A Corpus of Turkish Offensive Language. LREC 2020.

The dataset we use; provides labels but no fine-tuning-vs-prompting comparison.

Zampieri et al. (2023). OffensEval 2023: Offensive Language Identification in the Age of Large Language Models. Natural Language Engineering.

Zero-shot LLMs; Turkish only Flan-T5-large (0.451). No examples and no threshold calibration.

Dehghan & Yanıkoğlu (2024). Evaluating ChatGPT's Ability to Detect Hate Speech in Turkish Tweets. CASE 2024, pp. 54–59.

ChatGPT with 7 examples vs fine-tuned BERTurk on one Turkish hate dataset; one training size.

Pecher, Srba & Bielikova (2025). Comparing Specialised Small and General Large Language Models on Text Classification. EMNLP 2025, pp. 165–184.

English break-even near 100 labels; we test a noisy Turkish two-class task with run variance.

Schmitt & Schweter (2025). SindBERT, arXiv:2510.21364.

Turkish encoders on OffensEval-TR with full data only; no LLMs or small label budgets.

Şeker, Uludoğan, Önal & Özgür (2025). HATECAT-TR: A Turkish Hate Speech Dataset. Findings of EMNLP 2025.

Turkish span-level hate; BERTurk 0.59 vs LLaMA-3.3-70B 0.53. We classify whole tweets and score probabilities.

Karakaş & Şimşek (2026). arXiv:2606.29614.

Zero-shot LLMs vs BERTurk on Turkish sentiment at one size; we add examples, calibration and a label curve.

5 · DATA

Source: OffensEval-TR 2020, "A Corpus of Turkish Offensive Language" (Çöltekin, LREC 2020), https://coltekin.github.io/offensive-turkish/. Licence: annotations CC BY 2.0. Language: Turkish tweets; @USER/URL already masked in the release.

Split	Tweets	OFF	OFF %
Official train	31,756	6,131	19.3
Our train	30,000	5,792	19.3
Dev	1,756	339	19.3
Test (frozen)	3,528	716	20.3

After preprocessing: only the released @USER/URL masking; no lowercasing or deletion. 9 duplicate texts in train; 1 test/train overlap. Example (OFF, id 33518):

"Kırmızı ağır karar Allah belanı versin"

"A red card is a harsh call, God damn you."

No labelled data of our own; no agreement study needed.

Fields 6–9: method, protocol, results so far, error analysis

6 · METHOD AND BASELINE

Baseline as run: TF-IDF word 1–2-grams + char_wb 2–5-grams, min_df=2, sublinear, Turkish-aware lowercasing; logistic regression C=1.0, class_weight=balanced, max_iter=5000. Test 0.763 [0.745, 0.779].

Method implemented: BERTurk/XLM-R fine-tuning and Qwen/Gemma 4-bit prompting; Turkish instructions, A/B options, 8 examples (4+4), score log P(A) − log P(B); threshold maximises macro-F1 on the other 742 labels. One change vs baseline: labels train weights vs choose examples and a threshold.

Compute used: 2.4 GPU-hours fine-tuning (46 runs) and 2.5 L4-hours for 16 LLM passes; about 18 GPU-hours needed for the final (proposal), free Colab only.

7 · EVALUATION PROTOCOL

Macro-F1; frozen 3,528-tweet test; 10 BERTurk runs at 250/750/2,000 and 3 at 6,000/30,000; 3 LLM runs at 750. Paired bootstrap, 10,000 resamples: a gap is real only if BERTurk wins on ≥95% and the mean gap is ≥0.02. Fixed before method results; dev is for code checks only.

8 · RESULTS SO FAR (MEAN ± SD; BERTURK-30K = 0.832 ± 0.003)

System	Dev	Test
Always NOT	0.447	0.444
TF-IDF, 750	0.613 ± 0.015	0.620 ± 0.012
TF-IDF, 30,000	0.763 [0.739, 0.787]	0.763 [0.745, 0.779]
Qwen3-8B zero-shot	0.162	—
Qwen3-8B calibrated	0.628 ± 0.034	—
Qwen3-14B zero-shot	0.171	—
Qwen3-14B calibrated	0.773 ± 0.009	—

System	Dev	Test
Gemma-2-9B zero-shot	0.412	—
Gemma-2-9B calibrated	0.500 ± 0.048	—
Gemma-4-12B zero-shot	0.763	—
Gemma-4-12B calibrated	0.790 ± 0.007	—
BERTurk, 750	0.767 ± 0.014	—
XLM-R, 750 (Abl. A)	0.670 ± 0.025	—

9 · ERROR ANALYSIS — 60 AUDITED OF 275 DEV ERRORS, ONE EXAMPLE PER CATEGORY

False alarms (166)	n	Example
No insult / spurious cue	9	12420 · fan poll; no insult
Political/religious criticism	10	35124 · “Hırsız… İmamoğlu”
Sports criticism/banter	5	13105 · “padişahlık zannediyor”
Personal criticism	3	22409 · “fake hesapları sevmiyorum”
Sarcasm/irony	1	28737 · “Ha bi sen eksiktin”
Label noise/ambiguous	2	33218 · “Seni cehennem paklar”

Missed offensive (109)	n	Example
Implicit insult	8	43325 · “parazit canlı türü yok”
Masked profanity	5	15069 · “…kaldı amkk”
Veiled threat/violence	3	35789 · “idam edicem”
Political/religious insult	4	25929 · “Gerici yobazlar”
Institutional/sports criticism	3	20333 · “ÖSYM… güvenmeyebilir”
Label noise / not offensive	7	10001 · “veteriner okumaya…”

Fields 10–12: research lead, plan and risk, reproducibility

10 · RESEARCH LEAD SECTION — VESILE HAN (CSE 654), FINE-TUNING SIDE; GÜRKAN GÖZTEPELI (CSE 654), PROMPTING SIDE

Vesile Han. (1) Pecher, Srba & Bielikova, EMNLP 2025: small fine-tuned models catch up with general LLMs after about 100 labels on average, but need more on two-class tasks and once run variance is counted; all eight datasets are English. We test Turkish, a suffix-heavy language, on a noisy two-class task with 19.3% offensive tweets. (2) Schmitt & Schweter, SindBERT, arXiv:2510.21364 (2025): compare Turkish encoders on OffensEval-TR only with full training data, no LLMs and no small sizes. We add both. Ablation A (done on dev): BERTurk vs multilingual XLM-R-base at 750 labels, 10 runs each; BERTurk − XLM-R = +0.097 [0.080, 0.116], 100% bootstrap wins. Published comparison: KUISAIL reports plain fine-tuned BERTurk at 0.816 on this test set; our 30,000-label run should reproduce it.

Gürkan Göztepeli. (1) Şeker, Uludoğan, Önal & Özgür, HATECAT-TR, Findings of EMNLP 2025: fine-tuned BERTurk (0.59 F1) beat LLaMA-3.3-70B with a few examples (0.53) on Turkish hate-span detection; one training set, span labels. We classify whole tweets, vary the label budget, and read probabilities so every answer is valid. (2) Karakaş & Şimşek, arXiv:2606.29614 (2026): zero-shot open LLMs vs fine-tuned BERTurk on Turkish review sentiment at one size. We add examples, threshold calibration and a label curve. Ablation B: Gemma-2-9B with English instructions, same examples and threshold method, 3 runs.

11 · PLAN AND RISK

Done: BERTurk 750 (10) + 30k (3) and Ablation A XLM-R (10). 21–24 Nov: Ablation B English instructions. 25–27 Nov: zero-shot + contamination check. 28–30 Nov: freeze methods, open test, final runs. 1–5 Dec: bootstrap, error analysis, paper. 8 Dec: final report. Owners to be confirmed.

Biggest risk: a 26B+ model may not fit the L4; Gemma-4-26B-A4B OOM'd at 22 GB, so we used the 12B dense model. Fallback: Qwen3-14B alone. Result risk: the lines may never cross; the 750-label answer still stands. Contamination: give each LLM the first 200 test tweets and report exact completions. Proposal feedback: rubric asked for a test-set baseline; we scored the deterministic baseline once and kept methods on dev.

Ablation A: BERTurk 0.767 vs XLM-R 0.670; gap +0.097, 95% CI [0.080, 0.116], BERTurk wins on 100% of resamples.

12 · REPRODUCIBILITY

Repository: github.com/paysis/offenseval-tr-midproject (code, splits, saved predictions, results JSON). One command: bash scripts/reproduce.sh gpu. Seeds: split 42, subsets 42+k, bootstrap 42. Versions: Python 3.13.16 on Colab (3.14.8 local), scikit-learn 1.9.1, transformers 5.22.1, torch 2.14.1, bitsandbytes 4-bit NF4.

Roles: not decided yet; the team will assign them. AI tools: a coding assistant was used for code, analysis and drafting; every number in this deck is read from the saved result files and was checked by the team.

CI: green on main; 6 tests pass. No secrets, data zip or model weights are tracked.